

MATH221: LINEAR ALGEBRA I

Fall 2026

Instructor: Changho Han
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Time: Tues/Thurs 3:00pm – 4:15pm
Place: Hana Science Hall Basement 214.

Office Hours: TBA in Asan Science Hall 619.

Course Overview: This undergraduate course is a 1st part of a two-semester sequence in *linear algebra*. This is a hybrid proof-computation course, but this section is for non-math majors. In this course, you will mainly learn the foundational theory of linear algebra (with some abstraction), with some applications:

1. (Abstract) Vector Spaces, Vectors, and Linear Transformations: what do they mean?
2. Bases and Dimension of Vector Spaces: what do they mean and how to compute them?
3. Matrices: what do they mean and how to manipulate them? How are they related to linear transformations?
4. System of Linear Equations in Several Variables: how to solve them?
5. Determinants: what do they mean and how to compute them?

If you are just into learning computational aspects (without proofs) of this course, I **strongly recommend** looking elsewhere (outside of math department) for a course with similar content.

Prerequisites:

- Some familiarity with Korean High-School Math (basics of differentiation and integration over $\mathbb{R}$). Note that Calculus I (MATH161) is not required, but contains way more than all of what you need.

Textbook: Linear Algebra (5th ed) by Stephen H. Friedberg, Arnold J. Insel, and Lawrence E. Spence.

Grade Distribution: Attendance (10%), Quizzes (20%), Midterm (35%), Final Exam (35%).

$$F, D, D+ < 60 \leq C, C+ < 70 \leq B, B+ < 85, A, A+ \leq 100 \text{ (in \%)}$$

Note:

- There will be 2 Quizzes in Thursdays of weeks 5 and 13.
- There will be in-person Midterm and Final exams in Tuesdays of Week 8 and 16 respectively. This schedule is tentative and is subject to change as the semester progresses.
- Attendance starts with 10 points (out of 10), and 1 point is deducted whenever you miss a class. By the university policy, missing 10 (or more) classes will result an automatic F. In other words, make sure that your attendance grade is above 0 (missing at most 9 classes).

Tentative Course Outline:

- Weeks 1 - 4: §1 Vector Spaces
- Weeks 5 - 7: §2 Linear Transformations and Matrices (1): Linear Transformations and their Compositions, Kernel and Image, Isomorphisms.

- Week 8: Midterm
- Weeks 9 - 10: §2 Linear Transformations and Matrices (2): Matrix Representations of Linear Transformations, Algebra of Matrices, Change of Coordinates
- Weeks 11 - 13: §3 Elementary Matrix Operations and Systems of Linear Equations
- Weeks 14 - 15: §4 Determinants
- Week 16: Final Exam

Course Policies:

- This is an English course, i.e. everything will be in English.
- For an English course like this, your answers to every problem (both in quizzes and exams) must be in ENGLISH only! Writing any other language for your answers will result ZERO.
- When asking questions in class, please do your best to ask in English. If asked in Korean, I will repeat the question in English.
- The course website is in KU LMS(Learning Management System). The course website will include information about quizzes and grades.
- Some announcements will be in-class only, and some are in KU LMS. Students are responsible for keeping up with all the announcements (assignments, exam schedule) made in-class.
- The course will follow lecture notes that I will provide via KU LMS, instead of the textbook. The lecture notes, while based on the textbook, shuffles the order of materials from the textbook; and the lecture notes will use modern updated notations unlike the textbook. Please read those!
- Committing plagiarism is a serious violation of our school's code of academic conduct. Any student who commits plagiarism will immediately fail this course.
- Students with disabilities can request special accommodation related to attendance, lecture, assignments or tests through personal contact with a course professor at the beginning of semester.
- This syllabus is subject to change at the discretion of the instructor. Students are responsible for any changes that are announced in class.